

Final Machine Learning Project Report

Airbnb Price Prediction Using Regression Models

ITAI 1371 - Classic Machine Learning

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Final Project - August 2026

Dataset: Airbnb Open Data

Executive Summary

This project developed and compared regression models for predicting Airbnb listing prices. The final workflow used a corrected feature set after identifying target leakage, a required 70/15/15 train-validation-test split, five individual regression algorithms, an average ensemble of the best three validation models, and Bayesian Ridge. Random Forest Regressor was selected as the final model with test MAE 223.6204, MSE 76,230.0658, and R^2 0.3022.

1. Problem Definition and Dataset

The target variable is `price`, making this a regression problem. The project uses the Airbnb Open Data dataset from Kaggle. The cleaned project dataset contains 102,040 observations and 22 columns before final feature exclusions. The objective is to estimate listing price from available listing, location, review, availability, booking, and encoded categorical information.

Dataset source: <https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata>

2. Data Preparation

The cleaned midterm dataset was used as the starting point. Missing-value and duplicate checks were performed. The Boolean `instant\_bookable` variable was converted to numeric form. The target `price` was separated from the predictor matrix.

Target Leakage Correction

An initial run produced R^2 values near 0.995. Because this was unusually high for Airbnb price prediction, a sanity check was conducted. `service\_fee` had an approximately 0.998 correlation with `price`, and `fee\_to\_price\_ratio` was derived using price. Both were removed from the predictor set. Every model was retrained after the correction. This prevents the model from indirectly receiving the answer.

3. Train, Validation, and Test Methodology

The corrected data was split into 70% training, 15% validation, and 15% test sets using `random_state=42`. The training set was used to fit the models. The validation set was used to compare models and choose the top three for the ensemble. The test set was reserved for final evaluation. `StandardScaler` was fitted only on the training set and then applied to validation and test data for scale-sensitive models.

4. Models Trained

Linear Regression

Used as the linear baseline and trained on standardized features.

Decision Tree Regressor

A nonlinear tree model trained on the original feature scale.

Random Forest Regressor

An ensemble of 100 decision trees used to reduce single-tree instability.

Gradient Boosting Regressor

A sequential boosting model with 100 estimators and learning rate 0.1.

K-Nearest Neighbors Regressor

A distance-based regressor using five neighbors and standardized features.

Average Ensemble

The predictions of the top three validation models were averaged, as permitted by the project instructions.

Bayesian Ridge

A Bayesian linear regression approach used for the required Bayesian comparison.

5. Evaluation Metrics

MAE measures average absolute prediction error. MSE squares errors and therefore penalizes large misses more heavily. R^2 measures how much target variance is explained relative to predicting the mean. Higher R^2 and lower MAE/MSE are preferred. Negative R^2 indicates performance worse than the mean baseline.

6. Results

Model	Val MAE	Val MSE	Val R^2	Test MAE	Test MSE	Test R^2
Random Forest Regressor	224.8594	76,542.1309	0.3031	223.6204	76,230.0658	0.3022
Average Ensemble (Top 3)	265.0385	94,914.7498	0.1358	264.3793	94,434.8006	0.1356
Gradient Boosting Regressor	285.9776	109,639.0844	0.0017	285.4698	108,927.2936	0.0029
Bayesian Ridge	286.3809	109,854.4222	-0.0003	285.9414	109,253.1262	-0.0001
Linear Regression	286.4297	109,880.4135	-0.0005	285.9616	109,286.6749	-0.0004
K-Nearest Neighbors Regressor	301.4466	128,800.5200	-0.1728	302.7520	129,703.3286	-0.1873
Decision Tree Regressor	252.5198	144,582.2790	-0.3165	251.5091	144,107.3922	-0.3191

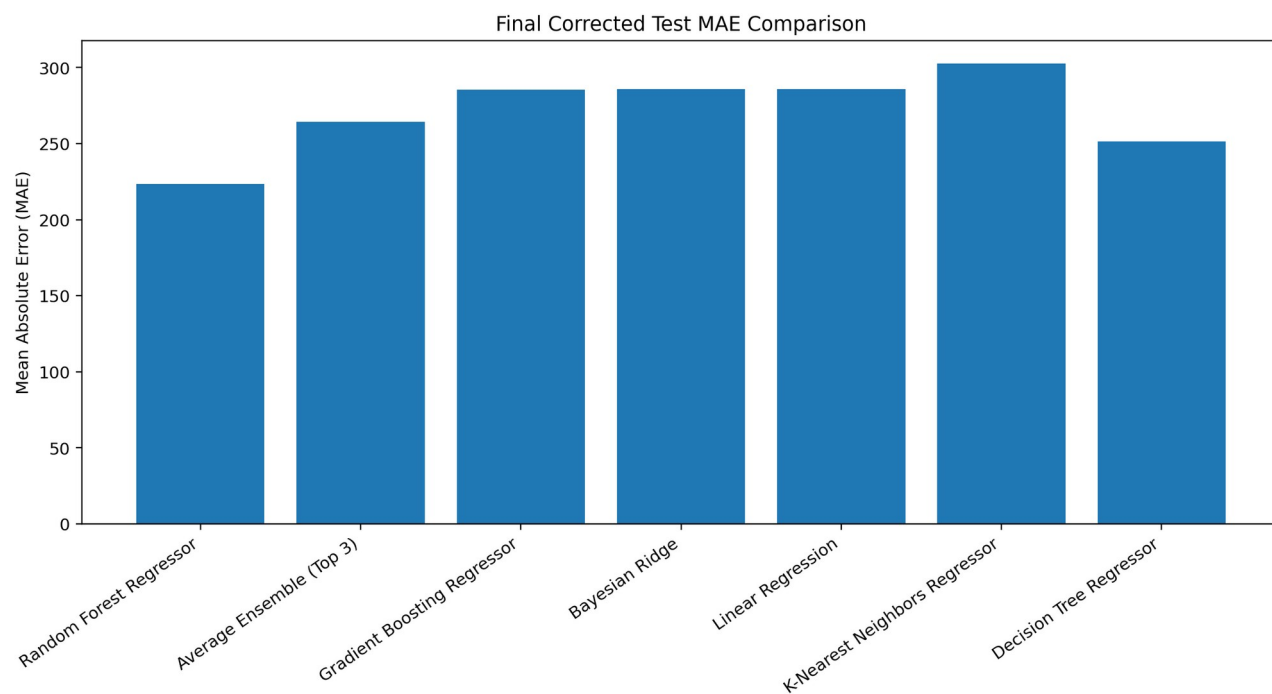


Figure 1. Final corrected test MAE comparison.

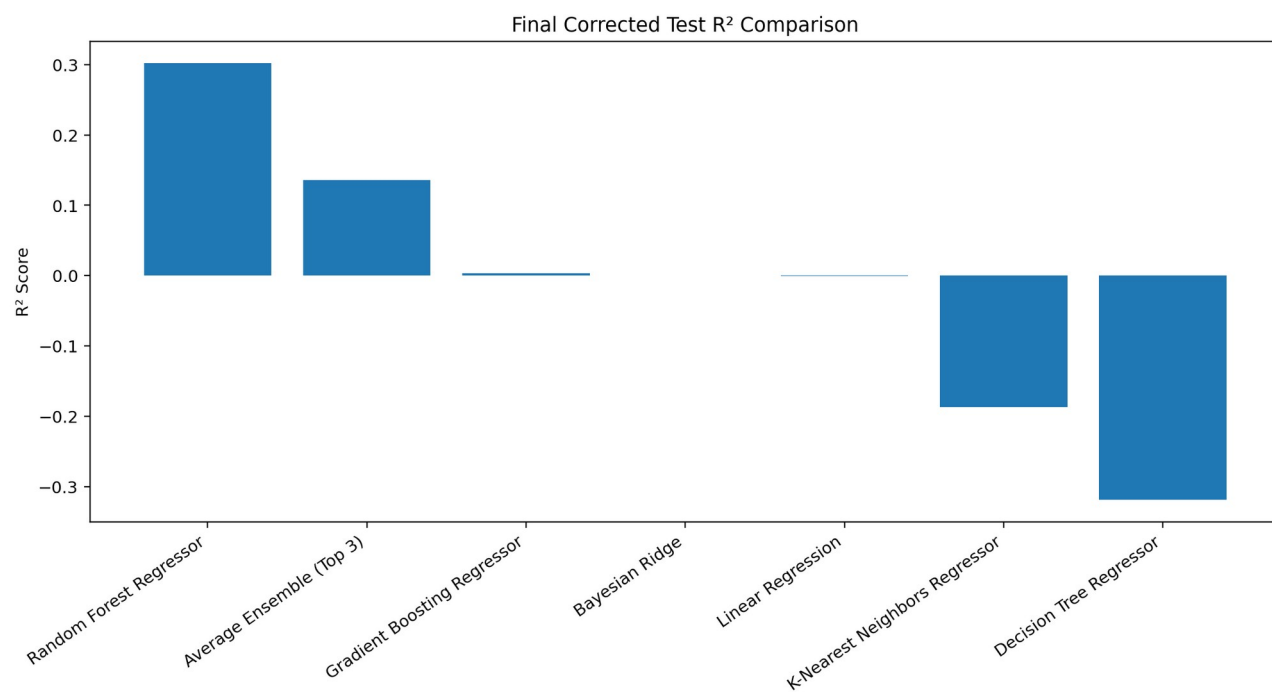


Figure 2. Final corrected test R^2 comparison.

7. Model-by-Model Analysis

Linear Regression

Linear Regression produced validation MAE 286.4297, MSE 109,880.4135, and R^2 -0.0005. Test R^2 was -0.0004. The near-zero R^2 shows that the remaining predictors did not explain Airbnb price well through a simple linear relationship.

Decision Tree Regressor

Decision Tree produced validation R^2 -0.3165 and test R^2 -0.3191. Its test MSE of 144,107.3922 was the highest of all evaluated approaches, indicating some large prediction errors and weak generalization.

Random Forest Regressor

Random Forest was the best model. It achieved validation MAE 224.8594, MSE 76,542.1309, and R^2 0.3031; test MAE 223.6204, MSE 76,230.0658, and R^2 0.3022. The model likely benefited from combining many trees and capturing nonlinear interactions among location, review, availability, and listing characteristics.

Gradient Boosting Regressor

Gradient Boosting achieved validation R^2 0.0017 and test R^2 0.0029. It performed slightly above the mean baseline but explained very little price variation with the selected configuration.

K-Nearest Neighbors Regressor

KNN achieved test MAE 302.7520 and R^2 -0.1873. Its negative R^2 and highest test MAE suggest that listings close in standardized feature space were not necessarily close in price.

Average Ensemble (Top 3)

The three best validation models were Random Forest, Gradient Boosting, and Linear Regression. Averaging their predictions produced test R^2 0.1356. The ensemble outperformed several models but did not beat Random Forest because averaging in weaker predictions reduced the strongest model's advantage.

Bayesian Ridge

Bayesian Ridge achieved test MAE 285.9414 and R^2 -0.0001. It was weaker than the Average Ensemble and Random Forest, indicating that a regularized Bayesian linear model did not capture the remaining nonlinear price relationships effectively.

8. Feature Importance of the Winning Model

Random Forest feature importance showed that longitude and latitude were the strongest individual predictors, followed by reviews per month and availability. This supports the idea that price depends on nonlinear location and listing-behavior relationships rather than a simple linear equation.

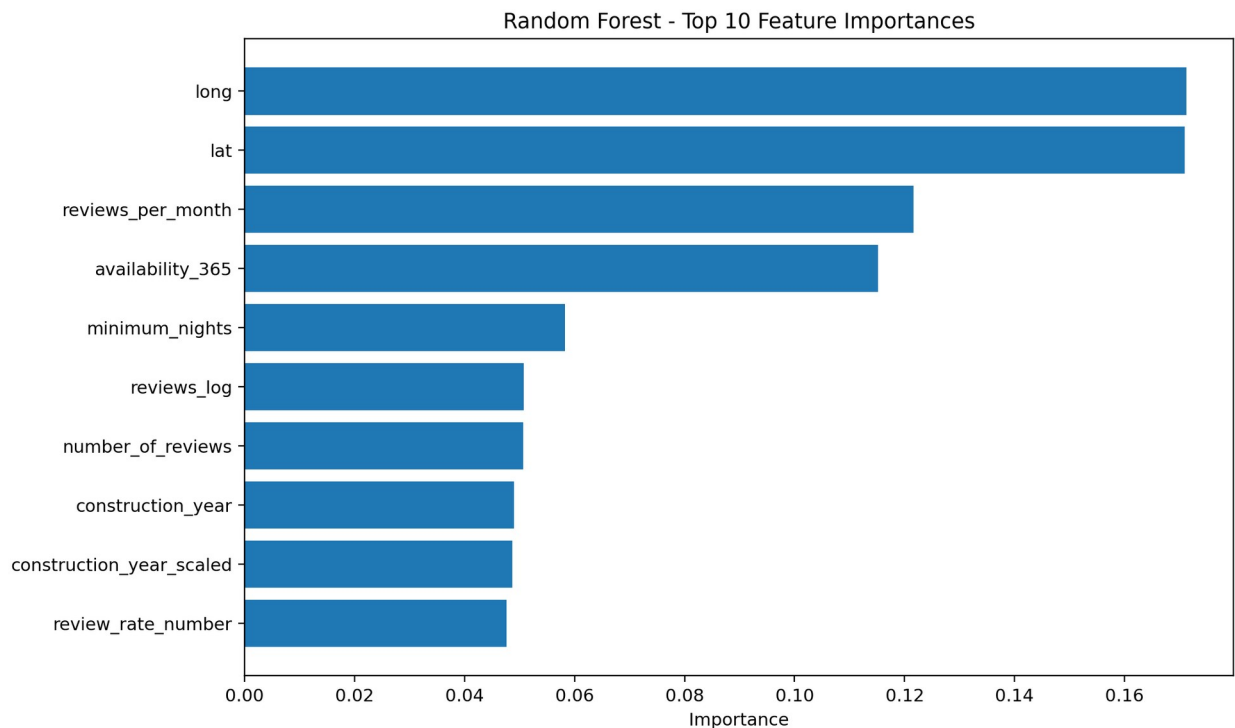


Figure 3. Top ten Random Forest feature importances.

9. Best Model and Why

Random Forest Regressor performed best because it achieved the highest test R^2 (0.3022), lowest test MAE (223.6204), and lowest test MSE (76,230.0658). Its validation and test R^2 values were almost identical, showing stable performance on unseen data. The algorithm can model nonlinear relationships and interactions without requiring the simple functional form assumed by linear models.

10. Limitations and Future Improvement

A test R^2 of 0.3022 means a large share of price variation remains unexplained. Future work could use richer neighborhood features, distance to landmarks, property amenities, text-based listing descriptions, seasonal information, or additional hyperparameter tuning. Cross-validation and models such as XGBoost or LightGBM could also be tested. These improvements should be performed while continuing to guard against target leakage.

11. Conclusion

The final corrected analysis selected Random Forest Regressor as the strongest model for Airbnb price prediction. The project satisfied the required model comparison, ensemble, Bayesian, validation, and test evaluation steps. The most important methodological lesson was that a very high score must be investigated before it is accepted. Removing price-derived variables changed

the results from an unrealistic R^2 near 0.995 to a more defensible test R^2 of 0.3022. The final results therefore reflect a more reliable estimate of generalization.